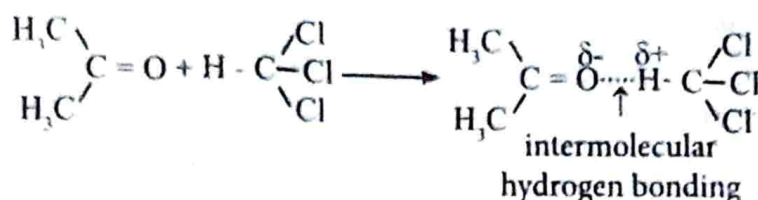


Experiment No. 10

Date: / /

Aim: To determine the enthalpy change during the interaction (hydrogen bond formation) between acetone and chloroform.

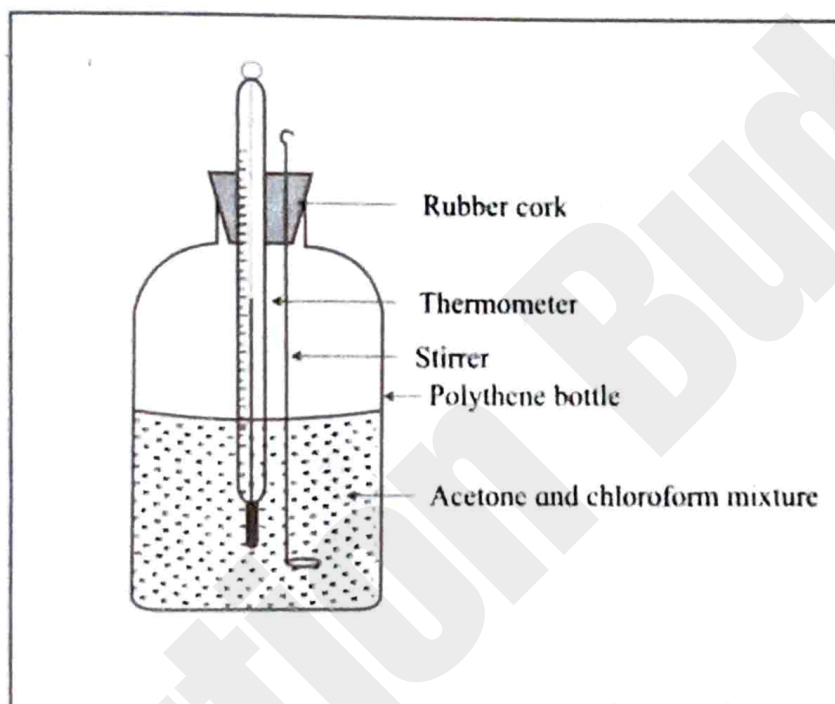
Theory: Intermolecular hydrogen bonding is formed, when acetone is mixed with chloroform. The process is exothermic and heat is evolved due to formation of intermolecular hydrogen bonding.



Apparatus: Polythene bottle, cork with two holes, stirrer, beakers, thermometer ($1/10^\circ$), measuring cylinder, etc.

Chemicals: Acetone, chloroform.

Diagram:



Procedure:

1. Take 25mL of acetone in dry polythene bottle.
2. Record its initial temperature ($t_1^\circ\text{C}$).
3. Take 25 mL of chloroform in a dry beaker.
4. Record its initial temperature ($t_2^\circ\text{C}$).
5. Add chloroform from beaker to the polythene bottle containing acetone, stir the mixture.
6. Record maximum (final) temperature of mixture ($t_3^\circ\text{C}$).

Note :

- i. Specific heat capacity of commonly used PET type of polythene bottle can be approximated to $1 \text{ J/g } ^\circ\text{C} = 1000 \text{ J/kg } ^\circ\text{C} \approx 0.25 \text{ cal/g } ^\circ\text{C}$
- ii. If the bottle is thin walled, its mass can be neglected if mass of water inside it is more than 100g.
- iii. If a heavy bottle of different material is used, we need to know and use its water equivalent.

Observation Table:

1	Initial temperature of acetone	t_1	22.°C
2	Initial temperature of chloroform	t_2	26.°C
3	Mean temperature acetone and chloroform	$t_3 = \frac{22(t_1) + 26(t_2)}{2}$	24.°C
4	Maximum temperature of mixture	t_4	37.°C
5	Rise in temperature	$\Delta t = 37^\circ\text{C} (t_4) - 24^\circ\text{C} (t_3)$	23.°C

Calculation: Given data

1	Density of chloroform	d_1	= 1.499 g/cm ³
2	Density of acetone	d_2	= 0.787 g/cm ³
3	Specific heat of chloroform	S_1	= 0.96 J/g °C
4	Specific heat of acetone	S_2	= 2.18 J/g °C

$$\begin{aligned}
 \text{Enthalpy change } (\Delta H) &= (25\text{mL} \times S_1 \times d_1 + 25\text{mL} \times S_2 \times d_2) \Delta t \\
 &= (25\text{mL} \times 0.96 \times 1.499 + 25\text{mL} \times 2.18 \times 0.78) \cdot 13 \Delta t \\
 &= (35.98 + 42.5) \times 13 \Delta t \\
 &= 78.48 \times 13 \Delta t = (z) \\
 &= (z) \times 4.184 \\
 &= 4268.8 \text{ J} \\
 &= 4.268 \text{ kJ}
 \end{aligned}$$

Space for log calculation

Number	log
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Result:

Enthalpy change due to formation of intermolecular hydrogen bonds when 25 mL of acetone is mixed with 25 mL of chloroform =kJ

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type questions.

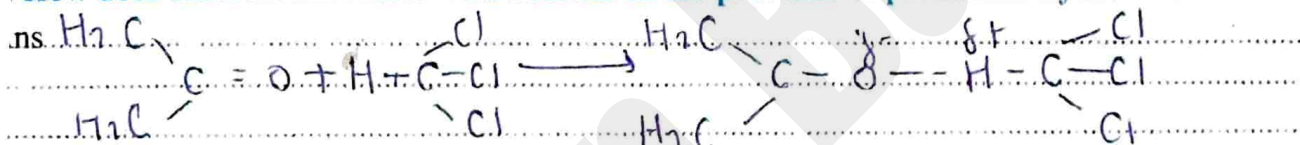
- The intermolecular hydrogen bonding present in between which atoms of chloroform and acetone
 - 'H' of acetone and 'Cl' of chloroform
 - 'H' of acetone and 'H' of chloroform
 - 'O' of acetone and 'H' of chloroform
 - 'O' of acetone and 'Cl' of chloroform
- The enthalpy change due to formation of intermolecular hydrogen bonding between chloroform and acetone is
 - exothermic reaction
 - endothermic reaction
 - reversible reaction
 - irreversible reaction
- The heat of formation of substance are
 - positive or negative
 - always negative
 - always positive
 - always zero
- A reaction in which heat is given off to the surroundings is said to be
 - exothermic reaction
 - endothermic reaction
 - reversible reaction
 - irreversible reaction

Short answer questions

1. What is intermolecular hydrogen bonding?

Ans. A hydrogen bond is an intermolecular force that forms a special type of dipole-dipole attraction when an hydrogen atom is bonded to a strongly electronegative atom in another molecule.

How does chloroform react with acetone in the presence of potassium hydroxide?



2. Write molecular formula of chloroform and acetone.

Ans. chloroform = CHCl_3
acetone = $\text{C}_3\text{H}_6\text{O}$

4. Write the name of functional group present in acetone.

Ans. Ketone - $\text{R}-\overset{\text{R}}{\text{C}}=\text{O}$

5. The intermolecular hydrogen bonding between chloroform and acetone is exothermic or endothermic reaction.

Ans. During the reaction between chloroform & acetone heat is evolved due to formation of intermolecular hydrogen bonding. so the reaction is an exothermic reaction.

Remark and sign of teacher: